



SWT 22022 – Practical for  
Internet Application  
Development

Department of Information  
and Communication  
Technology  
Faculty of Technology



**Lab sheet :13**  
**Reg. Number: SEU/IS/20/ICT/084**  
**Academic Year :2020/2021**  
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**Practical No :12**

# Title: Version Control with GitHub

## Aims:

- Setting up development environment
- Create repository
- Link changes to the repository

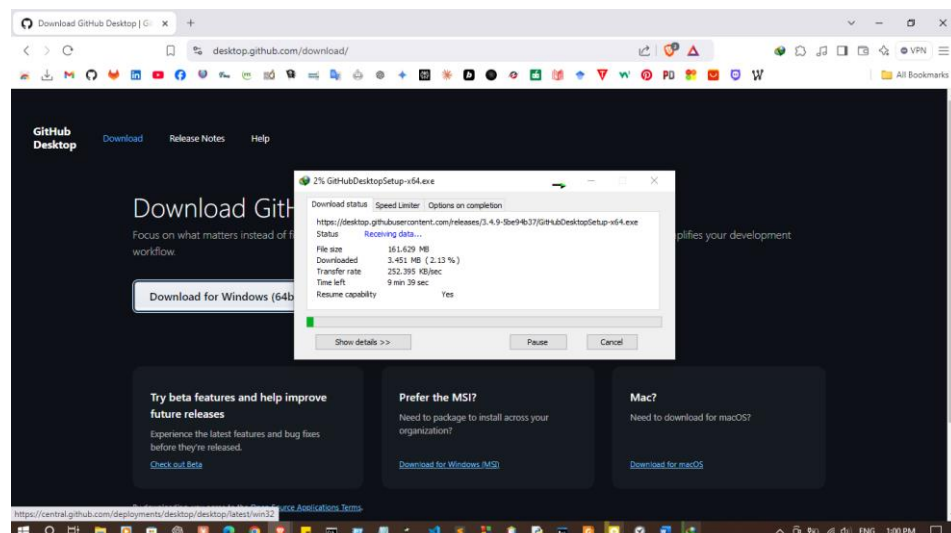
## Tasks:

- o Install Git on Windows
- o Configure Git
- o Create a GitHub Account
- o Create a Local Repository
- o Stage and Commit Changes
- o Create a Remote Repository on GitHub
- o Link Local Repository to GitHub
- o Push Changes to GitHub
- o Creating and Using Branches
- o Pull Changes to GitHub
- o Collaborating with GitHub

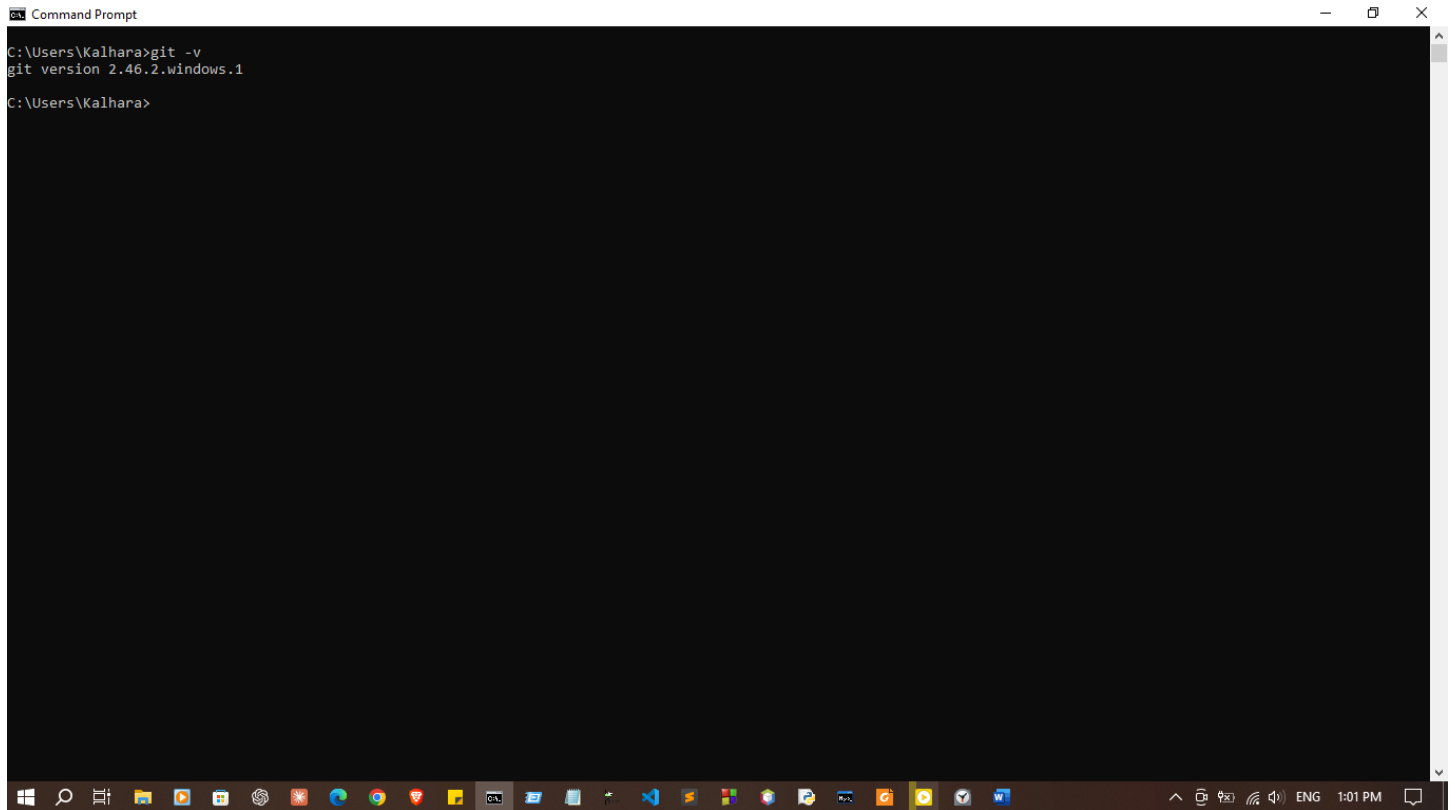
## Activities :

### 1. Install Git on Windows

- Download Git for Windows.



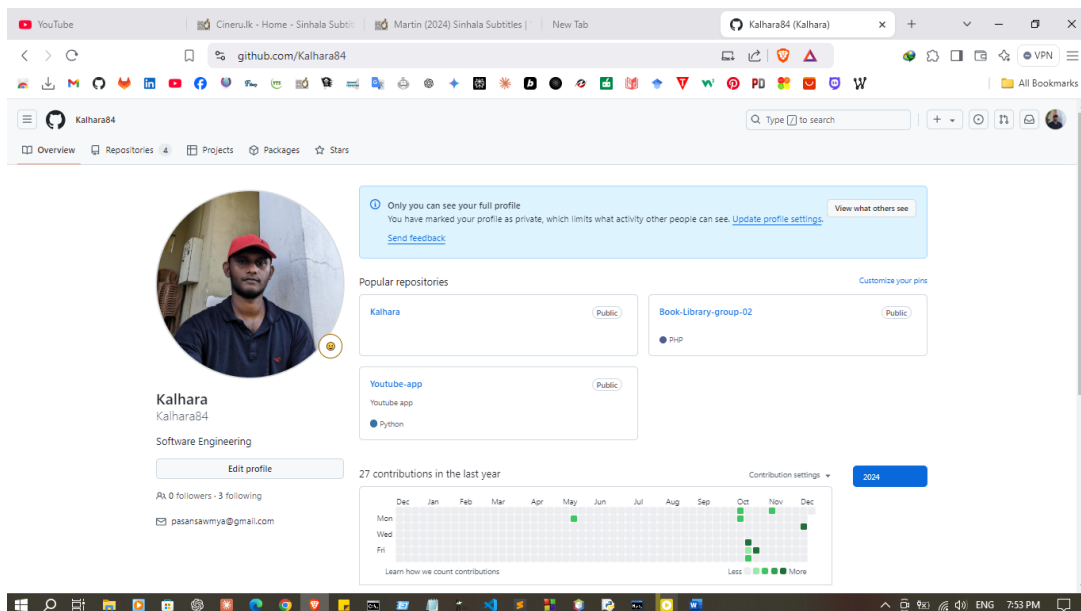
- Install Git and use the default settings.
- Verify the setup by running the following in the CMD
  - git --version



```
Command Prompt
C:\Users\Kalhara>git --v
git version 2.46.2.windows.1
C:\Users\Kalhara>
```

## 2. Create a GitHub Account

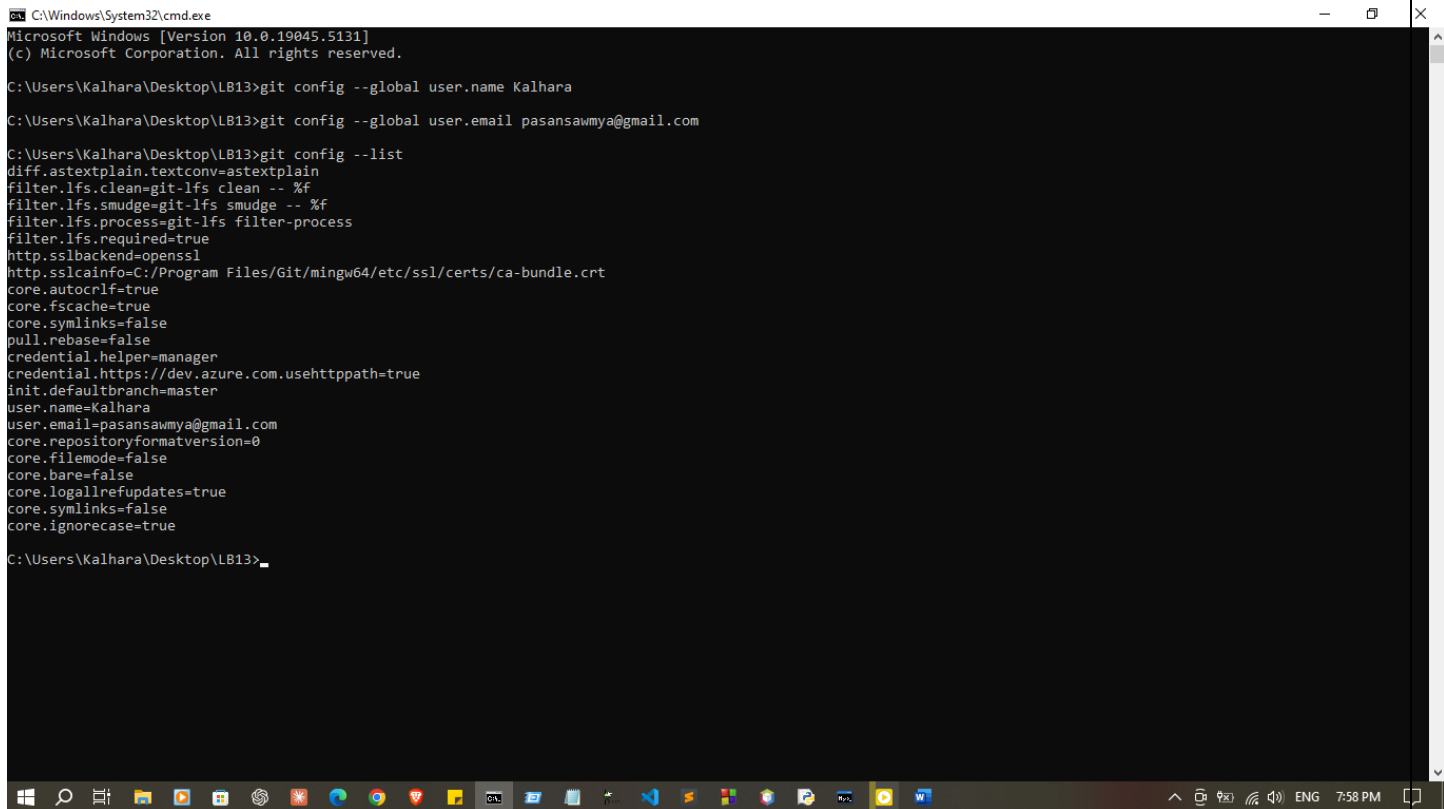
- Go to GitHub and Create an account



### 3. Configure Git

- Set up username and email for commits
- `git config --global user.name "Your name"`
- `git config --global user.email "email@example.com"`
- Verify the configuration ○ `git config --list`

```
git config --global user.name Kalhara
git config --global user.email pasansawmya@gmail.com
git config --list
```



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Kalhara\Desktop\LB13>git config --global user.name Kalhara

C:\Users\Kalhara\Desktop\LB13>git config --global user.email pasansawmya@gmail.com

C:\Users\Kalhara\Desktop\LB13>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Kalhara
user.email=pasansawmya@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true

C:\Users\Kalhara\Desktop\LB13>
```

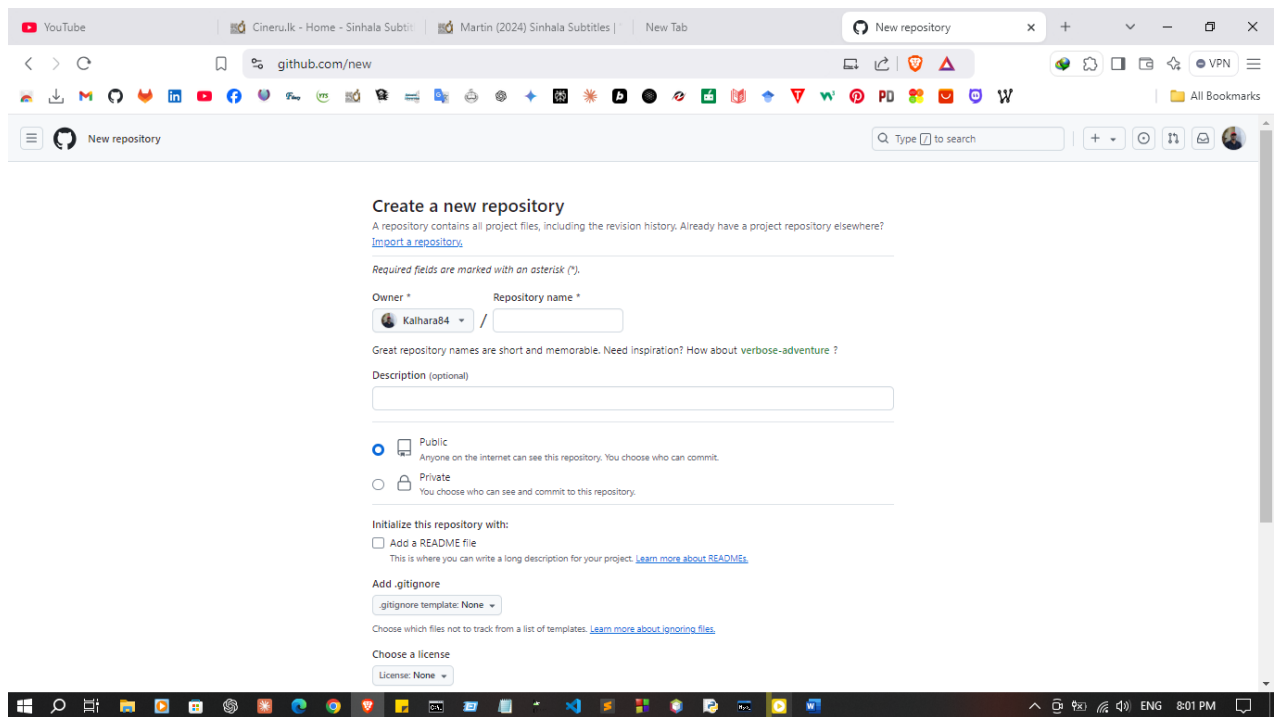
### 4. Create a Remote Repository on GitHub

- Log into GitHub and create a new repository

### 5. Create a Local Repository

- Open command prompt
- Navigate to your project directory
- Initialize a new Git repository

```
git init
git remote add origin https://github.com/Kalhara84/LB13.git
```



## Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere?

[Import a repository.](#)

*Required fields are marked with an asterisk (\*).*

Owner \*      Repository name \*

 Kalhara84 /

✓ LB13 is available.

Great repository names are short and memorable. Need inspiration? How about [verbose-adventure](#) ?

Description (optional)

- ☒  **Public**  
Anyone on the internet can see this repository. You choose who can commit.
- ☐  **Private**  
You choose who can see and commit to this repository.

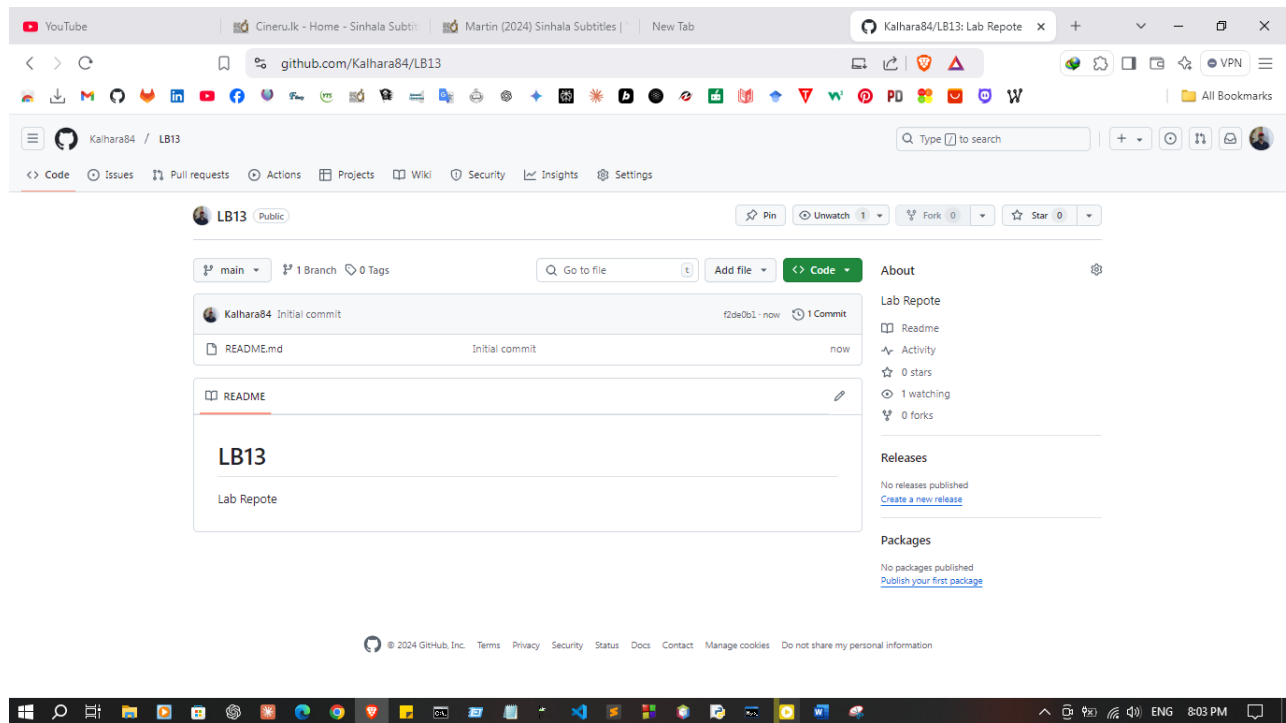
Initialize this repository with:

- ☒ **Add a README file**  
This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

Choose which files not to track from a list of templates. [Learn more about ignoring files.](#)

[Choose a license](#)



C:\Windows\System32\cmd.exe

```
C:\Users\Kalhara\Desktop\LB13>git init
Reinitialized existing Git repository in C:/Users/Kalhara/Desktop/LB13/.git/

C:\Users\Kalhara\Desktop\LB13>
```

C:\Windows\System32\cmd.exe

```
C:\Users\Kalhara\Desktop\LB13>git init
Reinitialized existing Git repository in C:/Users/Kalhara/Desktop/LB13/.git/

C:\Users\Kalhara\Desktop\LB13>git remote add origin https://github.com/Kalhara84/LB13.git

C:\Users\Kalhara\Desktop\LB13>
```

## 6. Stage and Commit Changes

- Add files to the staging area
  - git add . (for all files)
  - git add file\_name.txt (for one file)
- Commit the changes
  - Git commit -m "Initial Commit"

```
# Verify Git installation
git --version
# Configure Git
git config --global user.name Kalhara
git config --global user.email pasansawmya@gmail.com
git config --list
# Initialize a local repository
git init
# Add files to the staging area
git add .          # Add all files
git add text.html  # Add a specific file
# Commit changes
git commit -m "Add HTML File"
```

The screenshot shows the Visual Studio Code editor with a file named 'text.html' open. The file contains a simple HTML document with a header, main content, and footer. The terminal window on the right shows the execution of the following Git commands:

```
PS C:\Users\Kalhara\Desktop\LB13> git config --global user.name Kalhara
PS C:\Users\Kalhara\Desktop\LB13> git config --global user.email pasansawmya@gmail.com
PS C:\Users\Kalhara\Desktop\LB13> git init
Reinitialized existing Git repository in C:/Users/Kalhara/Desktop/LB13/.git/
PS C:\Users\Kalhara\Desktop\LB13> git add text.html
PS C:\Users\Kalhara\Desktop\LB13> git commit -m "Add HTML File"
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
PS C:\Users\Kalhara\Desktop\LB13>
```

## 7. Link Local Repository to GitHub

- Copy the repository URL
- Add the remote repository URL
  - git remote add origin paste\_url

**git remote add origin <https://github.com/Kalhara84/LB13.git>**

```
1 # Verify git installation
2 git --version
3
4 # Configure Git
5 git config --global user.name Kalhara
6 git config --global user.email pasansawmya@gmail.com
7 git config --list
8
9 # Initialize a local repository
10 git init
11
12 # Add files to the staging area
13 git add . # Add all files
14 git add text.html # Add a specific file
15
16 # Commit changes
17 git commit -m "Add HTML File"
18
19 # Add remote repository
20 git remote add origin https://github.com/Kalhara84/LB13.git
21
22 # Push changes to GitHub
23 git push -u origin master
24
25 # Create and switch branches
26 git checkout -b new-branch # Create and switch to a new branch
27 git checkout branch_name # Switch to an existing branch
28
29 # Push a branch to GitHub
30 git push -u origin new-branch
31
32 # Pull changes from the remote repository
33 git pull origin branch_name
```

```
PS C:\Users\Kalhara\Desktop\LB13> git config --global user.name Kalhara
PS C:\Users\Kalhara\Desktop\LB13> git config --global user.email pasansawmya@gmail.com
PS C:\Users\Kalhara\Desktop\LB13> git init
Reinitialized existing Git repository in C:/Users/Kalhara/Desktop/LB13/.git/
PS C:\Users\Kalhara\Desktop\LB13> git add text.html
PS C:\Users\Kalhara\Desktop\LB13> git commit -m "Add HTML File"
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
PS C:\Users\Kalhara\Desktop\LB13> git remote add origin https://github.com/Kalhara84/LB13.git
error: remote origin already exists.
PS C:\Users\Kalhara\Desktop\LB13> git push -u origin master
branch 'master' set up to track 'origin/master'.
Everything up-to-date
PS C:\Users\Kalhara\Desktop\LB13>
```

## 8. Push Changes to GitHub

- Push your commits to GitHub
  - `git push -u origin master`

**# Push changes to GitHub**  
**`git push -u origin master`**

```
PS C:\Users\Kalhara\Desktop\LB13> git config --global user.name Kalhara
PS C:\Users\Kalhara\Desktop\LB13> git config --global user.email pasansawmya@gmail.com
PS C:\Users\Kalhara\Desktop\LB13> git init
Reinitialized existing Git repository in C:/Users/Kalhara/Desktop/LB13/.git/
PS C:\Users\Kalhara\Desktop\LB13> git add text.html
PS C:\Users\Kalhara\Desktop\LB13> git commit -m "Add HTML File"
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
PS C:\Users\Kalhara\Desktop\LB13> git remote add origin https://github.com/Kalhara84/LB13.git
error: remote origin already exists.
PS C:\Users\Kalhara\Desktop\LB13> git push -u origin master
branch 'master' set up to track 'origin/master'.
Everything up-to-date
PS C:\Users\Kalhara\Desktop\LB13>
```

## 9. Creating and Using Branches

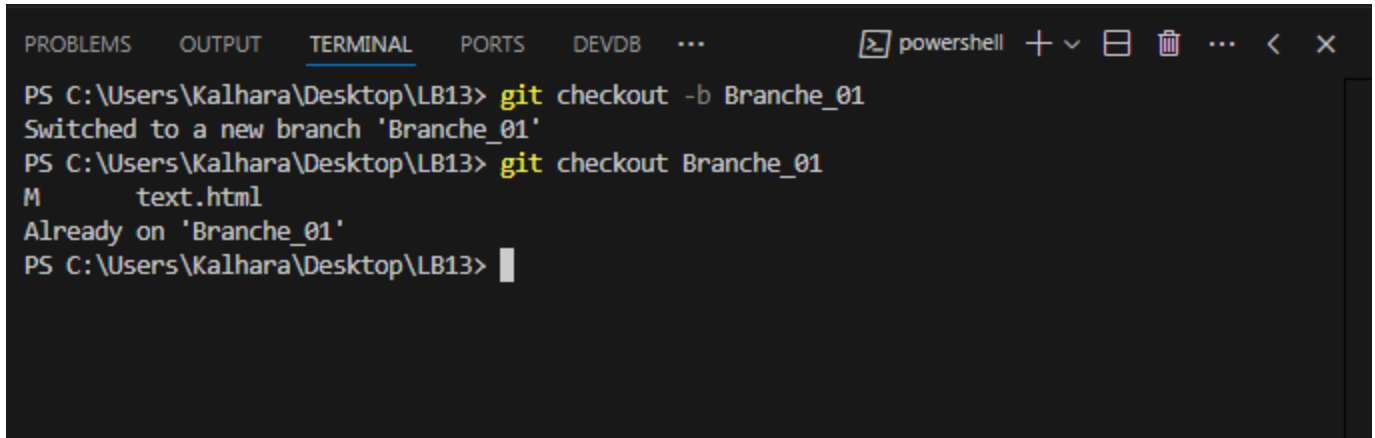
- Create a new branch
  - `git checkout -b new-branch`
- Switch to the branch

- git checkout new-branch
- Push the branch to GitHub

#### # Create and switch branches

git checkout -b Branche\_01 # Create and switch to a new branch

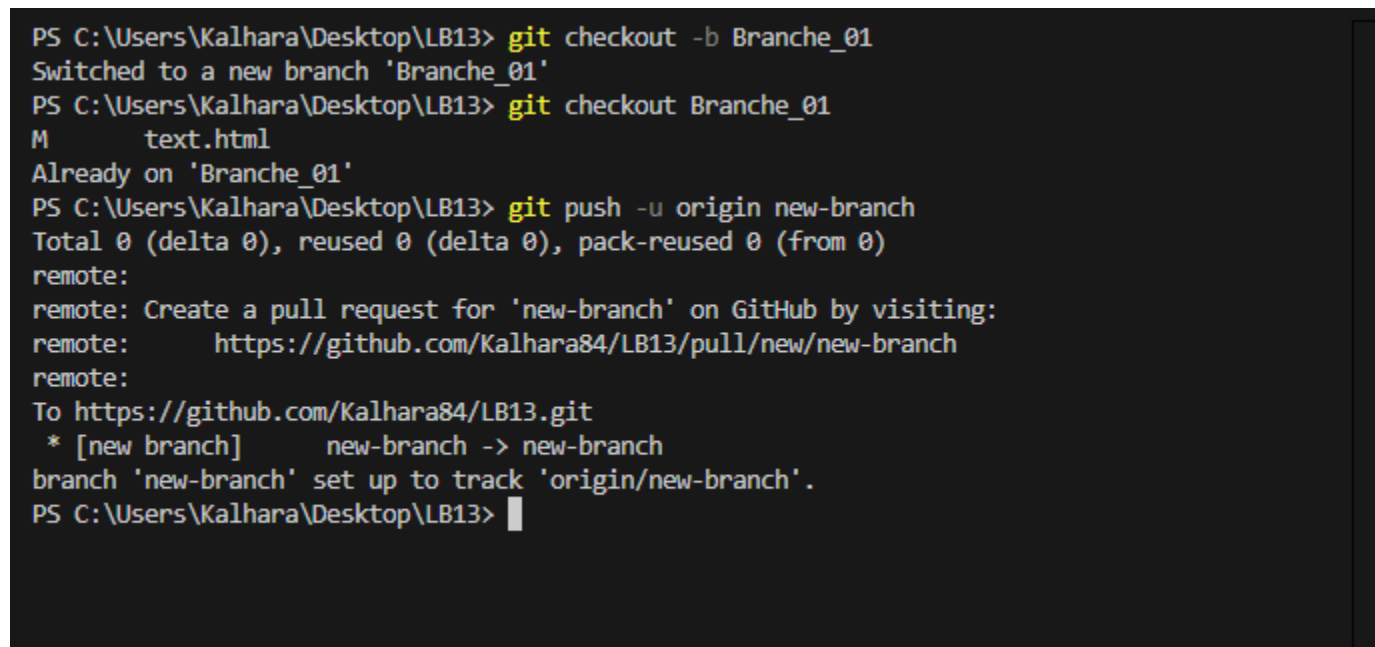
git checkout Branche\_01 # Switch to an existing branch



```
PROBLEMS OUTPUT TERMINAL PORTS DEVDB ... powershell + - [X]
PS C:\Users\Kalhara\Desktop\LB13> git checkout -b Branche_01
Switched to a new branch 'Branche_01'
PS C:\Users\Kalhara\Desktop\LB13> git checkout Branche_01
M      text.html
Already on 'Branche_01'
PS C:\Users\Kalhara\Desktop\LB13> 
```

#### # Push a branch to GitHub

git push -u origin new-branch



```
PS C:\Users\Kalhara\Desktop\LB13> git checkout -b Branche_01
Switched to a new branch 'Branche_01'
PS C:\Users\Kalhara\Desktop\LB13> git checkout Branche_01
M      text.html
Already on 'Branche_01'
PS C:\Users\Kalhara\Desktop\LB13> git push -u origin new-branch
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'new-branch' on GitHub by visiting:
remote:   https://github.com/Kalhara84/LB13/pull/new/new-branch
remote:
To https://github.com/Kalhara84/LB13.git
 * [new branch]      new-branch -> new-branch
branch 'new-branch' set up to track 'origin/new-branch'.
PS C:\Users\Kalhara\Desktop\LB13> 
```

|          |       |        |       |     |
|----------|-------|--------|-------|-----|
| Overview | Yours | Active | Stale | All |
|----------|-------|--------|-------|-----|

Q Search branches...

Default

| Branch | Updated    | Check status | Behind | Ahead   | Pull request |
|--------|------------|--------------|--------|---------|--------------|
| main   | 1 hour ago |              |        | Default | ...          |

Your branches

| Branch     | Updated        | Check status | Behind | Ahead | Pull request |
|------------|----------------|--------------|--------|-------|--------------|
| new-branch | 2 minutes ago  |              | 1      | 1     | ...          |
| master     | 26 minutes ago |              | 1      | 1     | ...          |

Active branches

| Branch     | Updated        | Check status | Behind | Ahead | Pull request |
|------------|----------------|--------------|--------|-------|--------------|
| new-branch | 2 minutes ago  |              | 1      | 1     | ...          |
| master     | 26 minutes ago |              | 1      | 1     | ...          |

## 10. Pull Changes from GitHub

- To fetch and merge changes from the remote repository

**# Push a branch to GitHub**

**git push -u origin Branche\_01**

**# Pull changes from the remote repository**

**git pull origin Branche\_01**

```
PS C:\Users\Kalhara\Desktop\LB13> git push -u origin Branche_01
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'Branche_01' on GitHub by visiting:
remote:   https://github.com/Kalhara84/LB13/pull/new/Branche_01
remote:
To https://github.com/Kalhara84/LB13.git
 * [new branch]      Branche_01 -> Branche_01
branch 'Branche_01' set up to track 'origin/Branche_01'.
PS C:\Users\Kalhara\Desktop\LB13> git pull origin Branche_01
From https://github.com/Kalhara84/LB13
 * branch            Branche_01 -> FETCH_HEAD
Already up to date.
PS C:\Users\Kalhara\Desktop\LB13> █
```

## 11. Collaborating with GitHub

- Clone the repository
  - Git clone repository\_url

# Clone a repository

git clone <https://github.com/Kalhara84/LB13.git>

```
PS C:\Users\Kalhara\Desktop\LB13> git push -u origin Branche_01
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'Branche_01' on GitHub by visiting:
remote:   https://github.com/Kalhara84/LB13/pull/new/Branche_01
remote:
To https://github.com/Kalhara84/LB13.git
 * [new branch]      Branche_01 -> Branche_01
branch 'Branche_01' set up to track 'origin/Branche_01'.
PS C:\Users\Kalhara\Desktop\LB13> git pull origin Branche_01
From https://github.com/Kalhara84/LB13
 * branch            Branche_01 -> FETCH_HEAD
Already up to date.
PS C:\Users\Kalhara\Desktop\LB13> git clone https://github.com/Kalhara84/LB13.git
Cloning into 'LB13'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
PS C:\Users\Kalhara\Desktop\LB13>
```

### Discussion:

- Version control with GitHub is an essential skill for managing code efficiently in a collaborative environment. In this lab practical, we learned to set up a development environment by installing and configuring Git on Windows. We started by creating a GitHub account and setting up a local repository to track changes. After staging and committing code, we established a connection between the local repository and a remote repository on GitHub. This allowed us to push changes online, ensuring our work was securely backed up and accessible to collaborators. We explored branch creation and usage for managing separate features or fixes, and practiced pulling changes to stay updated with others' work. The session highlighted the importance of collaboration by learning how to manage contributions from multiple users effectively on GitHub. This hands-on approach provided a comprehensive understanding of version control in a real-world development workflow.